

Class XII board exam data for 2024 show science is no longer a boys' club

Data on 2024 Class XII board exams, released by the Ministry of Education, **tell** a story of quiet **persistence**: For the first time in a **decade**, **more girls** — 28.14 lakh — **cleared** the board exams in the science stream than in **humanities** — 27.24 lakh. These figures **mark** a powerful shift in academic and social dynamics. As an analysis of numbers from 25 school boards between 2010 and 2023 by this newspaper in September 2024 showed, only 38.2 per cent of students clearing science in 2010 were girls. By 2023, that number had **climbed** to 45.5 per cent, **setting the stage for** the 2024 **breakthrough**. The **reversal** of the **long-standing** gender **tilt** toward humanities **is** an **affirmation** that **science** — once seen as a male **preserve** — **is** being actively **pursued** by female students across the country. In a country aspiring to be a global leader in the knowledge economy, this is welcome news: **Drawing on** the scientific **potential** of half its population **is** not just **desirable**, it is essential.

State-level results **underscore** the national picture. In Tamil Nadu, girls achieved a **formidable** 96.35 per cent science pass rate in 2024; in Jharkhand, Bihar and Odisha, girls **outperformed** boys. This **reshaping** of **ambition** and possibility **has** been **facilitated** by growing parental **encouragement** and good-faith policy interventions such as the Vigyan Jyoti scholarship schemes for girls from **disadvantaged** backgrounds, and the CBSE's Udaan initiative, that helps **underprivileged** girls prepare for engineering and medical entrance exams through mentoring and financial **aid**. What happens when more girls enter higher education in **STEM**? According to the All India Survey on Higher Education 2021-22, women now **make up** 52.1 per cent of all students in science **disciplines** across undergraduate, postgraduate, MPhil, and PhD levels. At the undergraduate level, they **constitute** a little over 51 per cent of the total enrolment. In medicine, women are **at par with** men; in engineering and technology — fields where they have traditionally been a **sliver** — their representation has risen **considerably**. These numbers **point to** a future workforce that is likely to be more **diverse**, representative, and **arguably**, more innovative.

The **promise** of these numbers, however, **will** continue to **come up against** the **shadow** of unequal realities unless it is met with deeper investment in infrastructure, mentorship networks, employment opportunities and safe spaces to learn and to **linger in**. **Girls** from rural areas and low-income households still **face immense** barriers that **cap** their ambitions: Lack of resources; unavailability of laboratories and equipment; a private study space, or simply, the right to stay in school. If this moment is to **make room** for long-term change, it must ensure that STEM dreams that have gotten **off the block** are not **stalled midway**.

[Practice Exercise]

- Red/blue coloring of words in the sentence indicates subject verb relationship; where 'red' denotes 'subject' and 'blue' denotes 'verb'.

Vocabulary

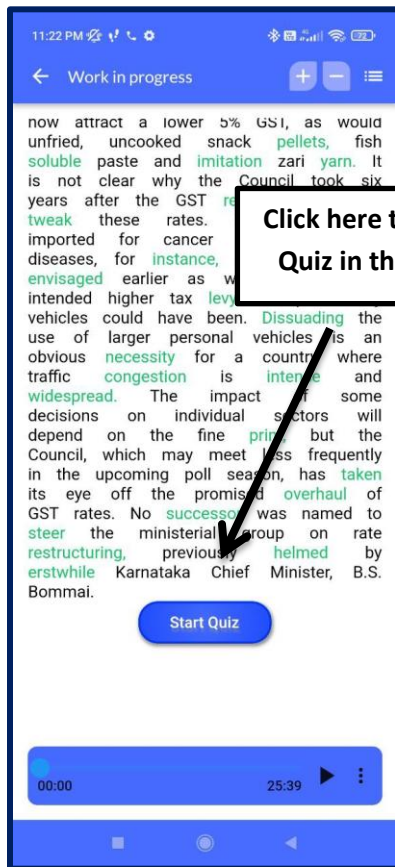
1. **Persistence** (noun) – determination, perseverance, tenacity, endurance, resolve
दृढ़ता
2. **Decade** (noun) – Period of ten years, दशक
3. **Humanity** (noun) – arts, liberal arts, social sciences, culture studies, philosophy मानवीय विषय
4. **Mark** (verb) – indicate, signal, denote, represent, signify चिन्हित करना
5. **Climb** (verb) – rise, increase, ascend, grow, surge बढ़ना
6. **Set the stage for** (phrase) – prepare for, lay groundwork, pave way for, enable, lead to मंच तैयार करना
7. **Breakthrough** (noun) – advancement, discovery, leap, milestone, progress उपलब्धि
8. **Reversal** (noun) – change, turnaround, flip, switch, undoing उलटफेर
9. **Long-standing** (adjective) – enduring, lasting, deep-rooted, established, historical लंबे समय से चला आ रहा
10. **Tilt** (verb) – lean, incline, skew, shift, favor झुकाव
11. **Affirmation** (noun) – confirmation, validation, endorsement, support, acknowledgment पुष्टि
12. **Preserve** (verb) – protect, maintain, safeguard, uphold, retain सुरक्षित रखना
13. **Pursue** (verb) – follow, chase, strive for, aim for, engage in पीछा करना
14. **Draw** (on) (verb) – utilize, use, rely on, tap into, depend upon उपयोग करना
15. **Potential** (noun) – ability, capacity, capability, promise, prospect संभाव्यता
16. **Desirable** (adjective) – preferable, ideal, beneficial, wanted, attractive वांछनीय
17. **Underscore** (verb) – highlight, emphasize, stress, point out, reinforce रेखांकित करना
18. **Formidable** (adjective) – impressive, powerful, strong, daunting, remarkable जबरदस्त / प्रभावशाली
19. **Outperform** (verb) – surpass, exceed, better, eclipse, outshine बेहतर प्रदर्शन करना
20. **Ambition** (noun) – goal, aspiration, dream, desire, objective महत्वाकांक्षा
21. **Facilitate** (verb) – enable, ease, assist, simplify, promote सुविधा देना
22. **Encouragement** (noun) – motivation, support, inspiration, boost, backing प्रोत्साहन
23. **Disadvantaged** (adjective) – underprivileged, deprived, marginalized, poor, backward वंचित

24. **Underprivileged** (adjective) – deprived, poor, disadvantaged, needy, marginalized पिछड़ा / गरीब
25. **Aid** (noun) – help, assistance, support, backing, relief सहायता
26. **STEM** (noun) – science, technology, engineering, mathematics विज्ञान, तकनीक, इंजीनियरिंग और गणित
27. **Make up** (phrasal verb) – constitute, comprise, form, account for, total बनना
28. **Discipline** (noun) – field, subject, domain, branch, area विषय
29. **Constitute** (verb) – comprise, form, compose, make up, establish बनना / स्थापित करना
30. **At par with** (phrase) – equal to, equivalent, on the same level, comparable to, matching बराबर
31. **Sliver** (noun) – fragment, slice, portion, trace, shred बहुत छोटा हिस्सा
32. **Considerably** (adverb) – significantly, greatly, markedly, notably, substantially काफी हद तक
33. **Point to** (phrasal verb) – indicate, suggest, reveal, show, demonstrate इंगित करना
34. **Diverse** (adjective) – varied, different, heterogeneous, inclusive, assorted विविध
35. **Arguably** (adverb) – possibly, potentially, debatably, conceivably, likely तर्क दिया जा सकता है कि
36. **Come up against** (phrase) – face, confront, encounter, oppose, run into सामना करना
37. **Shadow** (noun) – threat, looming danger, obstruction, hindrance, uncertainty बाधा / छाया
38. **Linger** (in) (verb) – remain, stay, hang around, persist, delay ठहरना / टिके रहना
39. **Immense** (adjective) – enormous, huge, massive, vast, tremendous अत्यधिक / विशाल
40. **Cap** (verb) – limit, restrict, ceiling, control, constrain सीमा तय करना
41. **Make room** (phrase) – allow for, accommodate, provide space, open up, enable स्थान बनाना
42. **Off the block** (phrase) – started, launched, initiated, got going, underway शुरुआत करना
43. **Stall** (verb) – stop, halt, delay, hinder, obstruct रोकना / अटकना
44. **Midway** (adjective) – half-done, partial, incomplete, in progress, in the middle बीच में / अधूरा

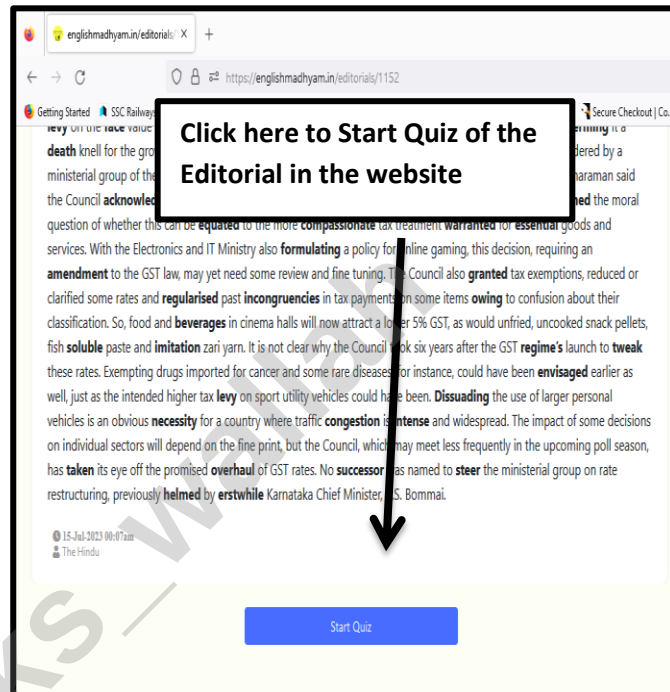
Summary of the Editorial

1. **Historic Milestone** – In 2024, more girls (28.14 lakh) than boys cleared the Class XII board exams in the science stream for the first time in a decade.
2. **Shift from Humanities** – More girls opted for and passed science than humanities (27.24 lakh), reversing a long-standing trend.
3. **Decade-Long Progress** – From just 38.2% in 2010, the percentage of girls passing science exams rose to 45.5% by 2023.
4. **Breaking Stereotypes** – Science, traditionally seen as a male domain, is now being embraced by female students nationwide.
5. **National Relevance** – Tapping the scientific potential of half the population is crucial for India's aspiration to lead the global knowledge economy.
6. **State-Wise Excellence** – States like Tamil Nadu (96.35% pass rate), Jharkhand, Bihar, and Odisha saw girls outperforming boys in science.
7. **Parental Support** – Rising family encouragement is helping girls pursue ambitions in science.
8. **Policy Push** – Government schemes like **Vigyan Jyoti** and **CBSE's Udaan** have supported girls from disadvantaged backgrounds.
9. **STEM Preparation** – These schemes provide mentoring and financial help to prepare girls for engineering and medical entrances.
10. **Higher Education Trends** – Women now account for **52.1%** of students in science disciplines across UG, PG, MPhil, and PhD levels.
11. **Undergraduate Majority** – At the UG level, women form just over **51%** of total enrolment in science.
12. **Medicine & Engineering** – Women are now at par with men in medicine, and their presence in engineering and technology is growing.
13. **Diverse Future Workforce** – This shift hints at a more diverse, inclusive, and potentially more innovative future workforce.
14. **Remaining Barriers** – Girls from rural and low-income backgrounds still face major hurdles like lack of labs, private study spaces, and basic rights to education.
15. **Call for Sustained Change** – To prevent drop-offs, deeper investments in infrastructure, mentorship, safety, and continued schooling are essential.

Practice Exercise: SSC Pattern Based



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